

ML Specification Report — playground-series-s3e5

Wine Quality (ordinal) Prediction · our from-scratch agent (GBDT pool + tree-search v2)

Figures grounded in the competition's `facts.json`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v2.json`.

Overview

The task is to predict wine quality on an ordinal 3–8 scale (metric: quadratic weighted kappa, QWK). We solve it with a from-scratch pipeline — a pool of gradient-boosted regressors (LightGBM / XGBoost / CatBoost) whose continuous output is discretized by an OptimizedRounder, grown through a multi-tier ablation and a 22-node tree search. Our benchmarked champion scores **CV QWK 0.5677** (exp #8, 6-way blend), climbing to 0.57066 once the tree search adds a boundary-push member. This beats the **NVIDIA reproduce-agent**, which copies Grandmaster `rsakata/optimize-qwk-by-lgb` verbatim and reaches only 0.5478 — our agent ahead by ~3.6%, a rare case where the from-scratch agent tops a Grandmaster kernel.

Why it matters. Automating sensory quality scoring from physicochemical measurements gives winemakers objective, scalable quality control; the ordinal target also makes this a clean testbed for QWK-optimized modelling.

Data

Purpose of Data. Predict the ordinal quality score (integers 3–8) of a wine from its physicochemical measurements — a Playground Series ordinal-regression episode scored by QWK. **Data Format** is clean, all-numeric **tabular CSV**: 11 continuous physicochemical features, no categorical columns. **Data Volume** is **2,056 training rows / 1,372 test rows**, with 11 raw predictive features once `Id` (a plain row index) and the target are removed.

Data Quality is high on the usual axes — **no missing values** in either split, **0 duplicate rows** (feature-only or full), and **no covariate shift** (largest train ↔ test mean difference 2.1%, on citric acid). The defining modelling wrinkle is **severe class imbalance**: quality 3 has only 12 rows (0.6%) and quality 8 only 39 rows (1.9%), while classes 5 and 6 dominate (839 + 778 = 79% combined) — an imbalance ratio of ~69.9×. The strongest EDA signals were:

- **Alcohol** is the dominant predictor (Spearman $r \approx +0.504$); **sulphates** next ($r \approx +0.457$).
- **Volatile acidity** ($r \approx -0.248$) and **total sulfur dioxide** ($r \approx -0.227$) are the strongest negative signals.
- **Mild collinearity** only: fixed acidity vs citric acid ($r \approx 0.696$), vs pH ($r \approx -0.674$), vs density ($r \approx 0.616$); free vs total SO_2 ($r \approx 0.638$) — both sides kept, as trees handle collinearity natively.
- **Modeling-head decision**: regression, not multiclass — QWK penalizes squared class distance, so a continuous regressor respects the ordering and lets the abundant mid-classes (5/6) inform placement

of the data-starved extremes (3/8). Reconfirmed this run: a diverse multiclass-EV member scored only 0.5204 and got zero blend weight.

Annotation Guidelines. The label is `quality` $\in \{3, 4, 5, 6, 7, 8\}$. Submissions are **integer quality predictions** scored by QWK — an agreement metric on discrete classes, so the continuous model output must be rounded to integers before scoring.

Feature Set. From the 11 raw columns we engineer **21 features** (11 $\rightarrow$ 21), all domain-informed ratios and interactions:

Group	Features
SO ₂ structure	<code>free_so2_ratio</code> , <code>bound_so2</code>
Acidity structure	<code>fixed_to_volatile_acid</code> , <code>citric_to_volatile_acid</code> , <code>total_acid</code> , <code>acid_ph_ratio</code>
Alcohol interactions	<code>alcohol_x_sulphates</code> , <code>alcohol_to_density</code> , <code>alcohol_x_density</code>
Fermentation	<code>sugar_to_alcohol</code>

Splitting strategy. A single canonical **5-fold StratifiedKFold on the raw `quality` label (seed = 42, shuffle)**, held fixed across all 12 experiments so scores are directly comparable. The leaderboard was **not attempted** (no Kaggle credentials on this machine), so every score here is **CV-only** — no Public/Private LB numbers exist.

Models & Architecture

Purpose of Architecture. An ordinal regressor that maximizes QWK on wine quality. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM / XGBoost / CatBoost regression members combined by a convex weighted blend, whose continuous output is mapped to integer classes by an OptimizedRounder; the benchmarked champion is a **6-way blend** (exp #8), and the tree search extends it to a **3-way blend** (node #17).

Each member takes the same **Input Format** — an all-numeric feature matrix (no categorical encoding needed) — of **Input Dimension 21 features** per row (a 1-D vector; no spatial/sequence structure).

Architecture Description. Each GBDT member is an additive ensemble of regression trees grown by histogram-based gradient boosting; the two strongest members (LGB_tuned, CAT_tuned) are produced by Optuna with the objective set directly to **post-rounder QWK on the trial's own OOF vector** (dodging the raw-vs-rounded discretization trap). The blend layer sits on top as a convex weighted average of members' out-of-fold predictions, with weights found by Dirichlet random search ($k = 800$) plus coordinate-ascent refinement, all scored on post-rounder QWK. **Model Complexity** is best expressed as trees $\times$ leaves rather than a dense parameter count — a tuned member's exact tree count is not recorded in the logs, and the champion blends up to 6 such members (the tree-search champion, 3).

Training procedures

Training proceeds as a **multi-tier ablation** on the shared 5-fold split, each tier measured on the same post-rounder OOF scale:

Tier	Configuration	OOF QWK
1	generic 3-model blend, raw 11 features, naive rounding (exp #1)	0.47871
2	21 features + regression blend + OptimizedRounder (exp #3)	0.52687
3	Dirichlet weight search + Optuna-tuned LGB & CAT pool, 6-way blend (exp #8)	0.56769
4	tree-search <code>harness_v2</code> (22 nodes, node #17 = 3-way blend)	0.57066

The **Loss Function** for each GBDT member is regression L2, but model selection and Optuna tuning are driven by the final **post-rounder QWK** (never the raw regression score — the s3e16 lesson). The **Optimization Algorithm** is gradient boosting (histogram-based) — **not** SGD/ADAM — with blend weights optimized by Dirichlet + coordinate-ascent search. The **Learning Rate** is the boosting shrinkage (LGB/CAT `learning_rate`), set per member by Optuna; the exact tuned value is *not recorded* in the logs. There is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

Training Duration for the tree search was **~697 s of wall-clock over 40 evaluated fits** (the v1 run; the v2 run was 793.7 s / 22 nodes); each Optuna tune was 40 trials / 600 s timeout, and the full 5-fold CV is cheap because the dataset is tiny (2,056 rows). **Training Memory** was not explicitly capped — a $2,056 \times 21$ float matrix is trivial on the single arm64 CPU machine; peak was *not recorded*. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-competition experience/prior injection** — distilled lessons (s3e14 keep-don't-replace, s3e16 decide-on-post-rounder) and the boundary-push prior (P18) that seeded the tree search's winning LGBBOUND member. **Data Augmentation** is N/A (*tabular*); the analogues are feature engineering (the 21 features) and seed-bagging (both tried; seed-bags gave zero weight here).

Reproducibility Standards are strict: fixed fold seed = 42, LightGBM determinism, and a resumable tree state (`experiments_tree_v2.json`) whose root and solo seeds reload from the v1 cache with bit-level agreement ($|\text{diff}| < 4\text{e-}6$, zero retraining). This being a native tree-search run, a **faithful replay is not needed** (the state file is the reproducible artifact).

Inference procedures

Decision Threshold is central here because QWK scores discrete classes: the regressor's continuous output is mapped to integers by an **OptimizedRounder** whose cutpoints are fit (Nelder-Mead) on the OOF vector to maximize post-rounder QWK, then applied with `clip[3, 8]`. The benchmarked champion (exp #8) uses cutpoints **[3.574, 4.586, 5.609, 6.174, 7.628]**; the tree-search champion (node #17) searches its cutpoints jointly with the blend weights inside each node's metric function (specific node-#17 cutpoint values *not recorded* as a static list). A nested (leave-fold-out) cutpoint diagnostic showed the full-OOF cutpoint overfit risk is modest and shrinking as the pool matured (gap 0.01644 on the 5-way blend $\rightarrow$ 0.00377 on the 6-way).

Inference Duration was not separately profiled; GBDT scoring of the 1,372 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Predict the integer quality class of each test wine. **Performance Metrics.** QWK is the sole competition metric; our benchmarked champion (exp #8) reaches **CV OOF QWK 0.5677**, along the progression $0.47871 \rightarrow 0.52687 \rightarrow 0.56769$, and the tree search adds a further **+0.00297 to 0.57066** (node #17). All figures are **CV-only** — the run was never submitted (no Kaggle credentials), so no Public/Private LB numbers exist.

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel verbatim:

Agent	Approach	CV QWK	Note
Our agent	from-scratch GBDT pool + tree-search	0.5677	winner
NVIDIA	reproduces <code>rsakata/optimize-qwk-by-lgb</code> (verbatim)	0.5478	-3.6%

Our agent **beats the NVIDIA reproduce-agent by +0.0199 (~3.6% relative)** — and does so against a kernel authored by a Kaggle **Grandmaster** (rsakata), which makes this one of the clearer wins for the originate-from-scratch approach over verbatim reproduction. The margin widens further (to 0.57066, ~+4.2%) once the tree search's boundary-push blend member is included. The honest caveats: this is a CV-only comparison (neither side has an LB number on this machine), and the champion's cutpoints are fit on full OOF — though the nested-cutpoint diagnostic bounds that overfit risk as small and shrinking.